PETRA ZLATARIĆ-NOVOSEL

Machine Learning Engineer

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PROFESSIONAL SUMMARY

Machine Learning Engineer with 5+ years of experience in natural language processing (NLP) and document classification, working from research and prototypes through production deployment. Hands-on experience fine-tuning transformer models, serving models with FastAPI and Docker on Azure, and monitoring production model quality. Additional experience classifying customer complaints in banking. Master's degree in computer science; PhD in progress focusing on Croatian-language named entity recognition.

TECHNICAL SKILLS

Languages and data: Python, SQL, pandas, NumPy
Machine learning and NLP: PyTorch, Hugging Face Transformers, scikit-learn, LightGBM, fastText, spaCy, BERTić, XLM-R, TF-IDF, feature engineering, model evaluation
Production ML and MLOps: MLflow, experiment tracking, model registry, FastAPI, Docker, Azure ML, Azure App Service, model serving, data drift monitoring
Engineering tools: Git, Linux

PROFESSIONAL EXPERIENCE

Machine Learning Engineer | Slavonica Document AI d.o.o.
03/2023 – Present

- Developed an automatic document classification model for public administration clients, achieving 94% accuracy across 23 classes, including invoices, contracts and administrative decisions.
- Fine-tuned transformer models (BERTić, XLM-R) for named entity recognition in Croatian documents.
- Introduced MLflow for experiment tracking and a model registry; models are served through FastAPI services in Docker containers on Azure ML and Azure App Service.
- Established production model quality monitoring, including data drift monitoring and weekly reporting.
- Mentor two student interns.

Data Scientist | Baranjska kreditna banka d.d.
09/2021 – 02/2023

- Built customer churn prediction models using gradient boosting and LightGBM for the sales team.
- Classified customer complaint topics using TF-IDF and logistic regression, later using fastText.
- Produced SQL reports from an Oracle data warehouse.

Research Project Assistant | Faculty of Applied Informatics, Sveučilište Drava
10/2020 – 08/2021

- Built and annotated a corpus of Croatian legal texts and developed Python scripts for the research project.

EDUCATION

PhD in Computer Science | Sveučilište Drava
2022 – Present | In progress
Research topic: Named entity recognition in Croatian legal texts.

Master's degree in Computer Science (Mag. ing. comp.) | Sveučilište Drava
2020

LANGUAGES

Croatian: Native | English: C1 | German: B2